



 Sustainable Freshwater Aquaculture

Aquaculture that empowers every farmer

A field-tested guide for over 3 crore small and marginal fish farmers — from hatchery operations and pond management to climate-resilient, solar-powered systems tested across Assam, Jharkhand and Odisha.

Explore Solar Solutions →



See How It Works

3Cr+

People dependent on fisheries¹

17MT

Annual fish production 2022-23²

2nd

India's global rank in production³

FIELD-TESTED ACROSS

Assam

Jharkhand

Odisha

ICAR–CMFRI

PMMSY

Why This Handbook Exists



Climate Disruption

Rising temperatures and erratic rainfall are disrupting breeding seasons and hatchery output nationwide.



Unreliable Power

Frequent grid outages force farmers toward costly, high-emission diesel generators.



Solar Solutions

Field pilots show 3× nursery survival and 20–54% lower operating costs with solar systems.



Real Farmer Income

Documented income gains from 17% to over 3,000% across different solar deployments.

"Farmers who've adopted solar-powered systems tell a consistent story: nursery survival rates climbing threefold, operational costs dropping by 20 to 54 percent, and incomes rising dramatically."

— India Fisheries Handbook, Executive Summary

Features

Achieve **farm clarity** with tools built to simplify every stage — from hatchery to market.

Everything a farmer needs, nothing they don't.

HANDBOOK COVERAGE

20 Chapters

Disease Checker & Water Quality

Spot symptoms early, get instant likely-cause guidance, and keep DO, pH and ammonia in the safe zone.

Climate-Resilient Tech

Real solar deployment data — RoI, capex and income impact — from six pilots across three states.

Profit Calculator & Records

Estimate yield and net profit before you stock, then log every harvest to catch losses early.

 Solar Powered

 Live Fish

Nursery Survival Rate

3× Higher

RoI 2.13

₹1,90,000 Capex

+

Start Building a Climate-Resilient Farm in 3 Easy Steps

1

Pick your species & system. Use the Species Explorer and Value Chain map to match fish and infrastructure to your water and climate.

2

Adopt climate-smart tech. Explore solar aeration, biofloc and RAS pilots with documented RoI from Assam, Jharkhand and Odisha.

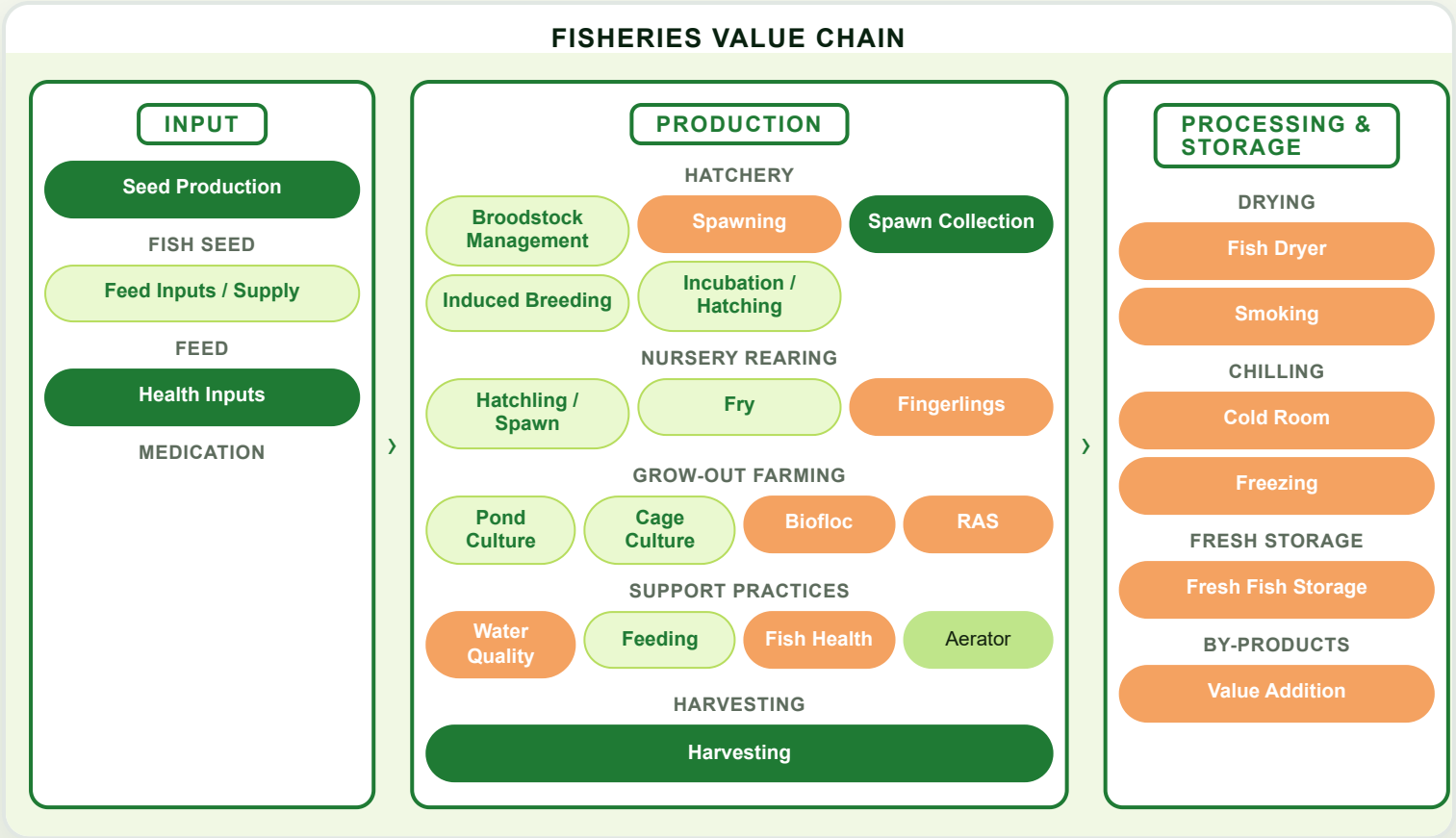
3

Track profit & records. Run the Profit Calculator before you stock, then log every harvest in the Record Keeper.

[Learn more about the Climate-Resilient Farming Pilot Program](#)

Fisheries Value Chain

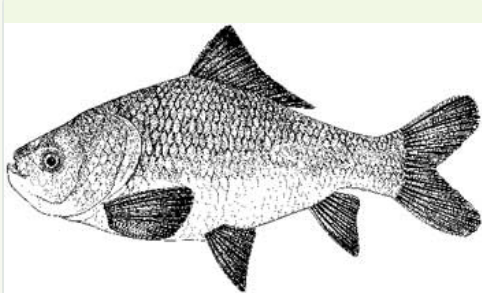
An interactive map of how fish move from seed to market. Click any node to learn what it involves, why it matters, and what challenges and solutions exist at each stage.



Species Photo Guide

Before choosing what to farm, understand what's available. This chapter introduces the key species — Indian Major Carps, exotic species, high-value indigenous fish, and small indigenous species — with real photographs and guidance on which suits your climate, water type, and market.

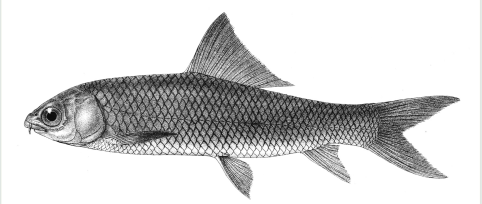
- All Species
- Indian Major Carps
- Exotic / Non-Carp
- High Value
- Small Indigenous



Catla
Catla catla
IMC



Rohu
Labeo rohita
IMC



Mrigal
Cirrhinus mrigala
IMC



Grass Carp

Ctenopharyngodon idella

CARP



Silver Carp

Hypophthalmichthys molitrix

CARP



Common Carp

Cyprinus carpio

CARP



Amur Carp

Cyprinus carpio haematopterus

SELECTIVE BREED



GIFT Tilapia

Oreochromis niloticus

EXOTIC



Pangasius

Pangasius hypophthalmus

EXOTIC



Magur

Clarias batrachus

HIGH VALUE



Murrel

Channa striata

HIGH VALUE



Pabda

Ompok bimaculatus

HIGH VALUE



Kawai / Climbing Perch

Anabas testudineus

HIGH VALUE



Singhi

Heteropneustes fossilis

HIGH VALUE



Mola

Amblyparyngodon mola

SIS

The following sections walk through each stage of the fisheries value chain — from hatchery operations and species selection to grow-out farming, processing, and marketing. Each chapter covers what the stage involves, the challenges farmers face, and the climate-smart solutions being deployed to overcome them. Click on any section in the sidebar or scroll through to explore.

Hatchery

The foundation of aquaculture. This chapter covers what hatcheries do, the challenges they face in rural India, and the climate-smart infrastructure being deployed to make seed production reliable and year-round.

 **Spawning, Incubation & Nursery Systems — What, Why, How** 

Biofloc Technology

A climate-smart grow-out solution. Biofloc eliminates water exchange, reduces feed costs by 30–40%, and turns waste into food — making it ideal for water-scarce and flood-prone regions.

 **Biofloc Technology — Zero Water Exchange Farming** 

Recirculating Aquaculture Systems (RAS)

The most advanced climate-resilient solution. RAS recycles 95–99% of water indoors, enabling year-round production independent of weather, grid power, or land availability.

 **Recirculating Aquaculture Systems (RAS) — The Most Intensive System** 

Water Quality Parameters

The invisible factor that determines everything. Learn the optimal ranges for temperature, oxygen, pH, ammonia, and more — and what to do when parameters go outside safe limits.

TEMPERATURE**25-32°C**

Below 20°C → slow growth

Optimal**DISSOLVED OXYGEN****> 5 mg/L**

Below 3 → stress; below 1 → mortality

Optimal**PH LEVEL****7.0-8.5**

Below 6 or above 9 → stress

Optimal**AMMONIA (NH₃)****< 0.02 mg/L**

Above 0.5 mg/L → toxic

Monitor**NITRITE (NO₂)****< 0.1 mg/L**

Above 1 mg/L → brown blood disease

Monitor**TRANSPARENCY****25-40 cm**

Below 15 → bloom; above 60 → low nutrients

Monitor**ALKALINITY****75-200 mg/L**

Stabilises pH; lime corrects low levels

Optimal**H₂S****0 mg/L**

Any detectable level = emergency

Danger

Disease Checker

Identify what's wrong with your fish. Select the symptoms you observe and get possible diagnoses with treatments — based on the handbook's disease tables. Always consult a fish pathologist for severe outbreaks.

Select all symptoms you are observing:☐ Fish surfacing / gasping at surface☐ White or grey spots on body/fins☐ Ulcers or red lesions on skin☐ Fins fraying, rotting or missing☐ Excessive mucus / slimy body☐ Rapid or sudden fish deaths☐ Bloated abdomen / fluid in cavity

☐ Fish rubbing against pond sides

☐ Refusal to eat / lethargy

☐ Whirling or erratic swimming

🐟 Get Diagnosis

Fish Processing, Solar Drying & Marketing

Post-harvest value creation. This chapter covers how fish are preserved, processed, and marketed — with solar-powered solutions that eliminate diesel dependency and extend shelf life from hours to months.



Fish Processing, Solar Drying & Marketing — From Water to Market



Climate-Resilient Technologies: Real Deployments

Proof that it works. Real photographs and verified ROI data from field deployments in Assam, Jharkhand, and Odisha — showing how solar-powered systems eliminate grid dependency and transform small-farmer economics.

NURSERY POND (ASSAM)

3x

Survival rate — spawn to fingerling
— with spiral diffuser aeration

INCOME INCREASE (ASSAM)

3179%

Kalong Kapili pilot. Capex Rs
1,90,000. RoI: 2.13

GROW-OUT AERATOR
(JHARKHAND)

46.5%

Income increase. Survival up 20%.
Capex Rs 5,80,000

COST REDUCTION

20-54%

Operational cost reduction across
all pilot systems

160W Diaphragm air blower — nursery
aeration system

Diffuser-type aeration in operation

Nursery aeration — Kalong Kapili,
Assam

Fou



Why Climate-Resilient?

In rural India, grid electricity is unreliable — frequent blackouts and voltage fluctuations force farmers to rely on costly diesel generators. Climate-resilient aquaculture systems **eliminate this constraint entirely**, enabling round-the-clock aeration and processing independent of the grid.

Farm Record Keeper

Track your farm's performance. Log harvests, revenue, and expenses — all saved locally in your browser. Export to CSV anytime for accounting or government scheme applications.

POND / TANK NO.

e.g. P1

DATE

07/24/2026



HARVEST QUANTITY (KG)

0

SELLING PRICE (RS/KG)

0

TOTAL EXPENSES (RS)

0

SPECIES

Rohu



+ Add Record

↓ Export CSV

🗑 Clear All

Date	Pond	Species	Qty (kg)	Price (Rs/kg)	Revenue (Rs)	Expenses (Rs)	Profit (Rs)
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No records yet.

Farm Profitability Calculator

Plan before you invest. Estimate expected yield, revenue, costs, and energy savings based on real pilot data from Assam, Jharkhand, and Odisha — across pond, cage, biofloc, and RAS systems.

POND AREA (HA)

0.5

FARMING SYSTEM

Pond Farming



PRIMARY SPECIES

Rohu / Catla (IMC)



CULTURE PERIOD (MONTHS)

6

SELLING PRICE (RS/KG)

120

SOLAR SYSTEM?

Yes — Climate Resilient



Calculate Expected Profit